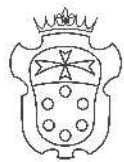
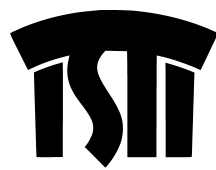


Understanding any Time Series Classifier with a Subsequence-based Explainer

Francesco Spinnato*, Riccardo Guidotti, Anna Monreale, Mirco Nanni, Dino Pedreschi, Fosca Giannotti

*francesco.spinnato@di.unipi.it
University of Pisa

ECML PKDD 2024 - Nectar Track



SCUOLA
NORMALE
SUPERIORE

Understanding any Time Series Classifier with a Subsequence-based Explainer

Francesco Spinnato*, Riccardo Guidotti, Anna Monreale, Mirco Nanni, Dino Pedreschi, Fosca Giannotti

*francesco.spinnato@di.unipi.it
University of Pisa

ECML PKDD 2024 - Nectar Track

Spinnato, Francesco, et al. "Understanding Any Time Series Classifier with a Subsequence-based Explainer." ACM Transactions on Knowledge Discovery from Data 18.2 (2023): 1-34.

Understanding any Time Series Classifier with a Subsequence-based Explainer

Francesco Spinnato*, Riccardo Guidotti, Anna Monreale, Mirco Nanni, Dino Pedreschi, Fosca Giannotti

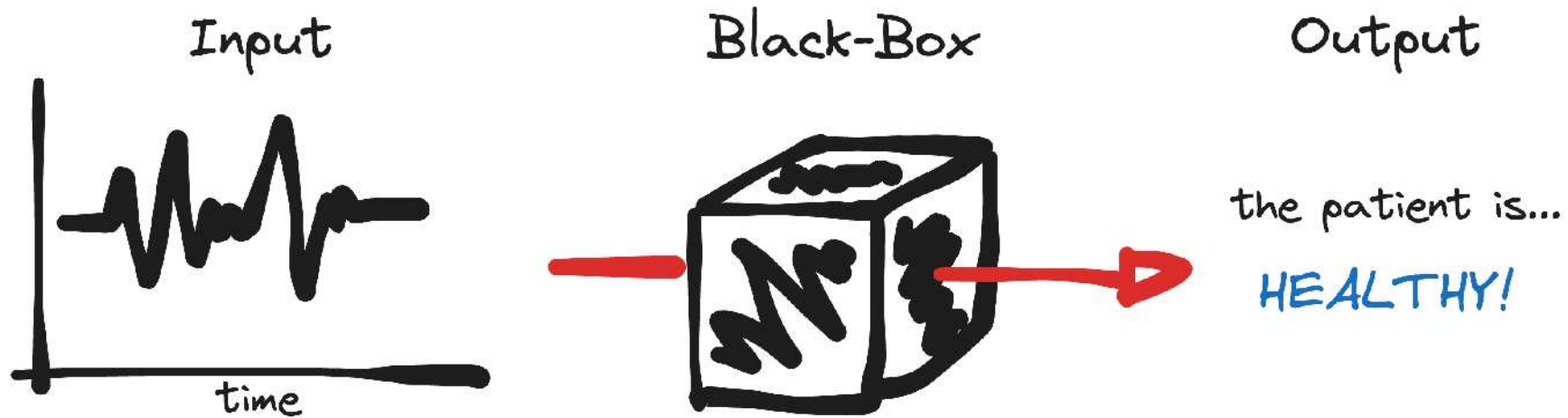
*francesco.spinnato@di.unipi.it
University of Pisa

ECML PKDD 2024 - Nectar Track

Extension of: Guidotti, Riccardo, et al. "Explaining any time series classifier." 2020 IEEE second international conference on cognitive machine intelligence (CogMI). IEEE, 2020.

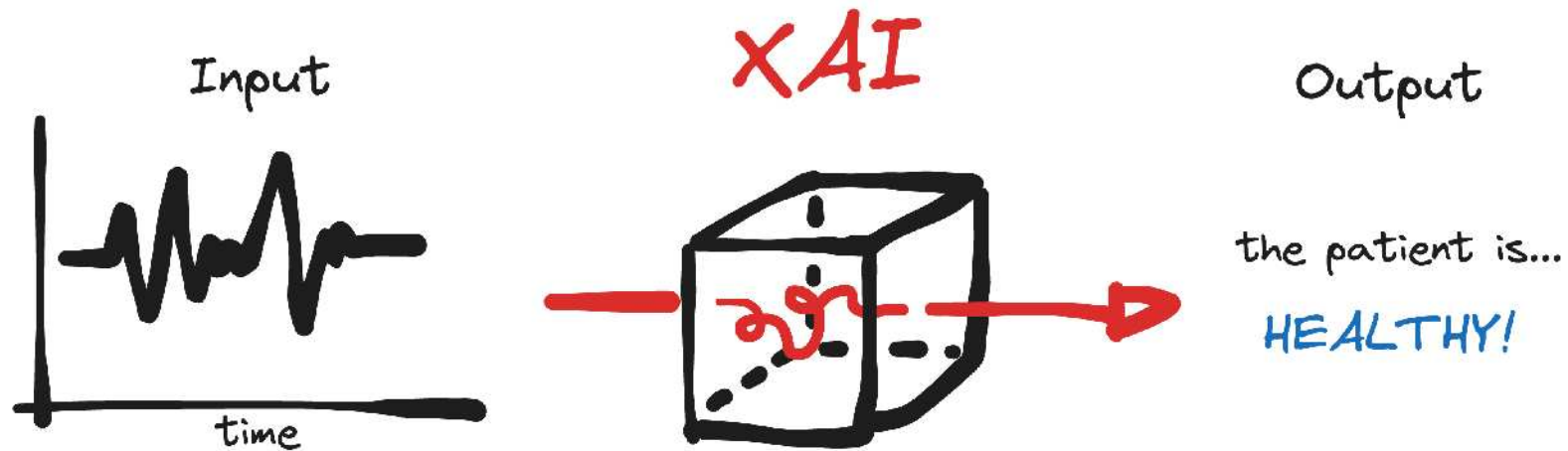
Time Series Classification

Time Series Classification (TSC) is the task of **predicting a single categorical output** given a whole time series.



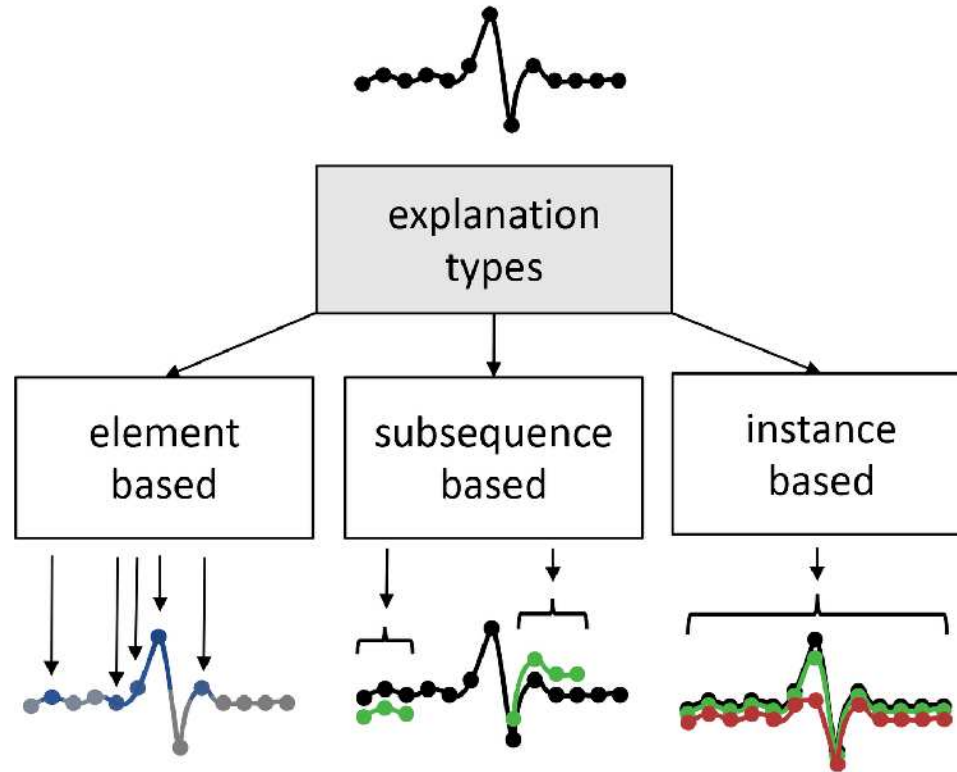
XAI for Time Series Classification

Explainable AI tries to understand the **relationship between the input ant the output.**



Explanation Types

There are many kinds of explanations for time series data.



The best TSC models are blackboxes, so XAI is fundamental but...

... usually XAI approaches for TSC have the following **limitations**:

- they output **one explanation kind**;
- they work on **univariate** time series data;
- they are tested on simple **'toy' datasets**.

Local Agnostic Subsequence-based Time Series explainer (LASTS)

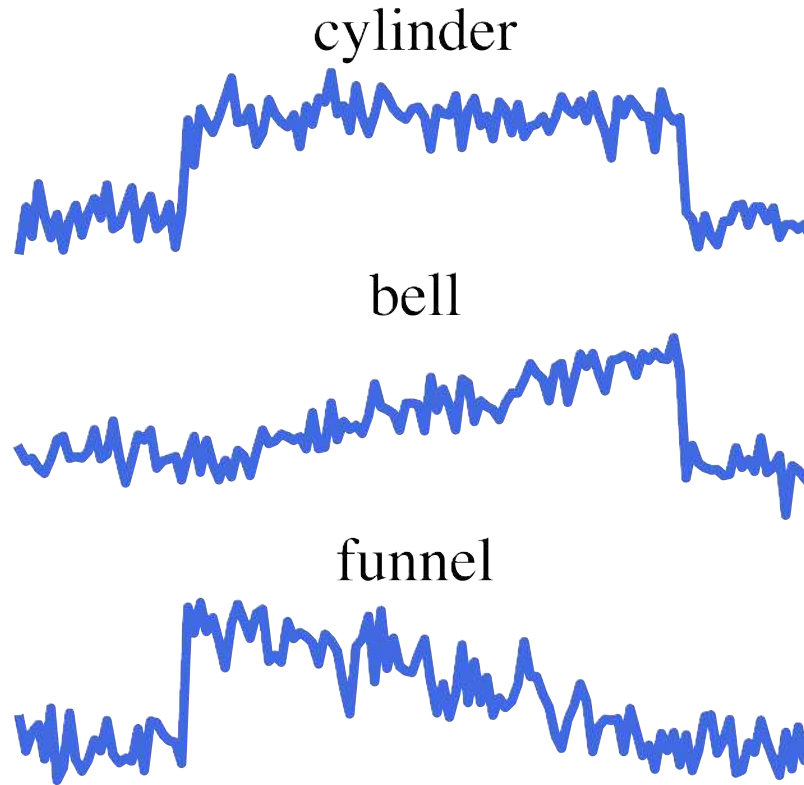
- **Local**: explains the prediction of the black-box for single univariate/multivariate instances (not the entire model);
- **Agnostic**: can explain any black-box.

LASTS can output three kinds of explanations:

- **Element-based**: saliency map;
- **Subsequence-based**: factual/counterfactual rules;
- **Instance-based**: prototypical/counterfactual instances.

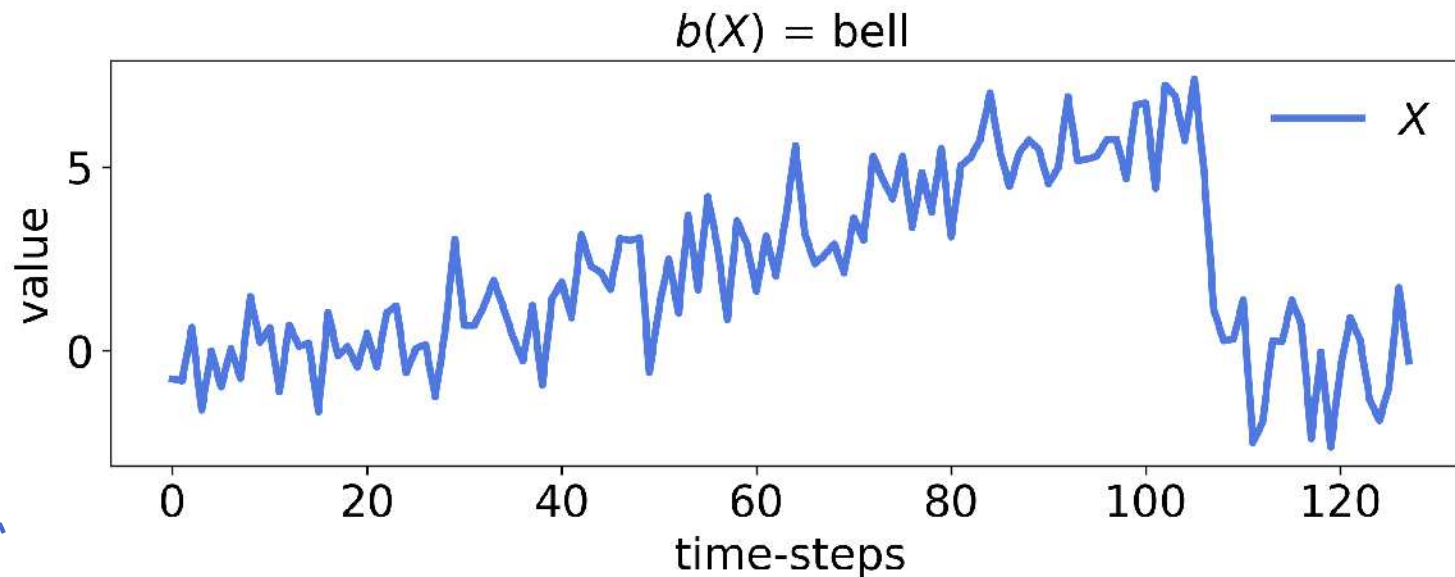
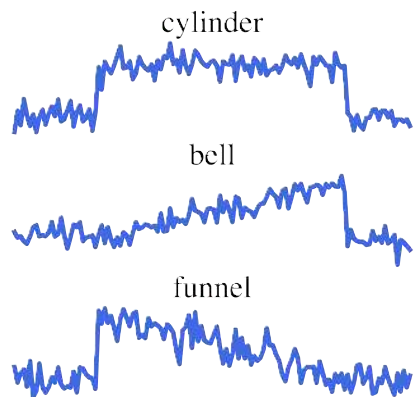
An Example of an Explanation

Let's take CBF, a simple dataset with **three classes**.



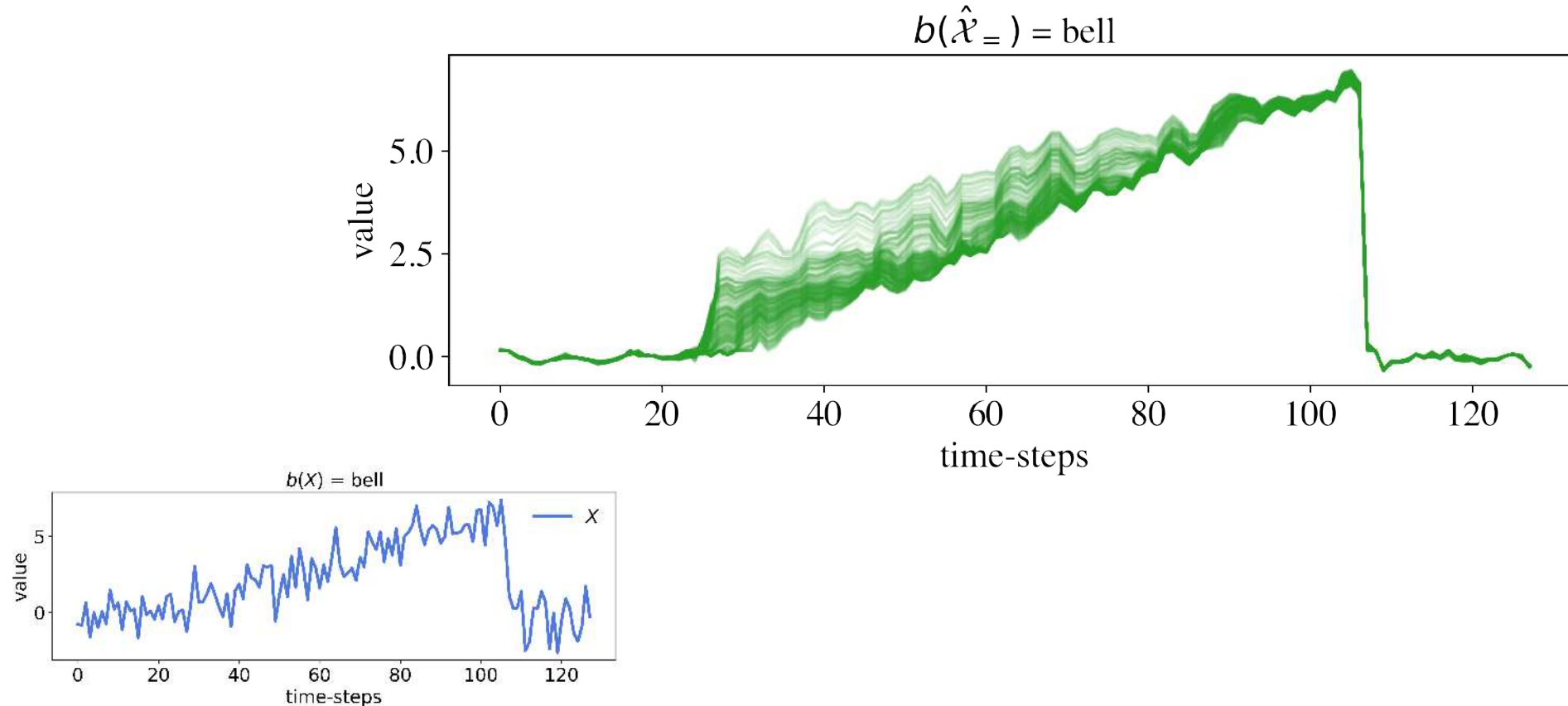
An Example of an Explanation

We want to **explain the prediction** of a black-box b for the time series X , i.e., why $b(X) = \text{bell}$?



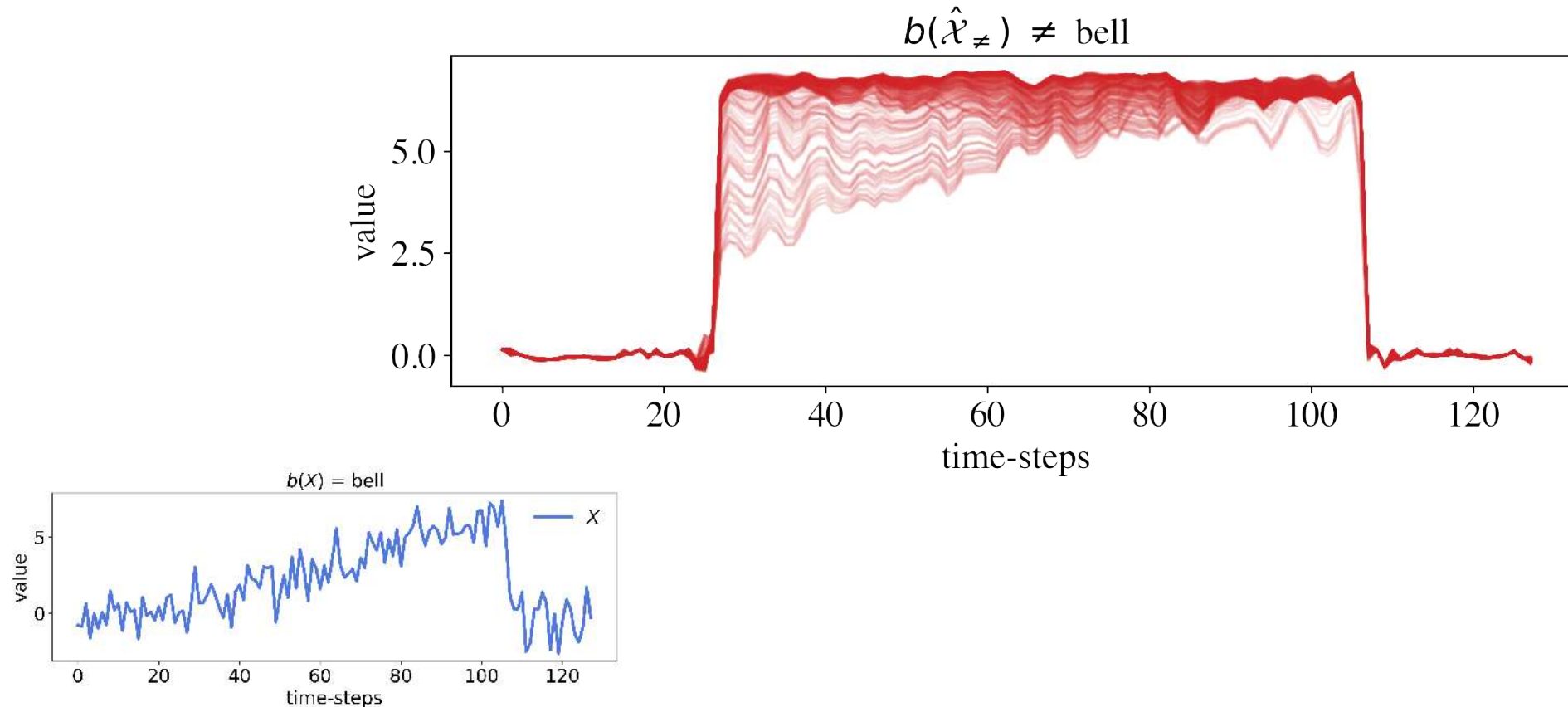
1. Instance-based Explanation: Prototypes

Prototypes are instances similar to X and with the same class.



1. Instance-based Explanation: Counterfactuals

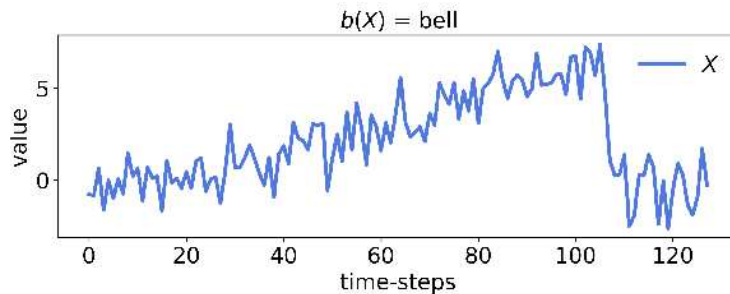
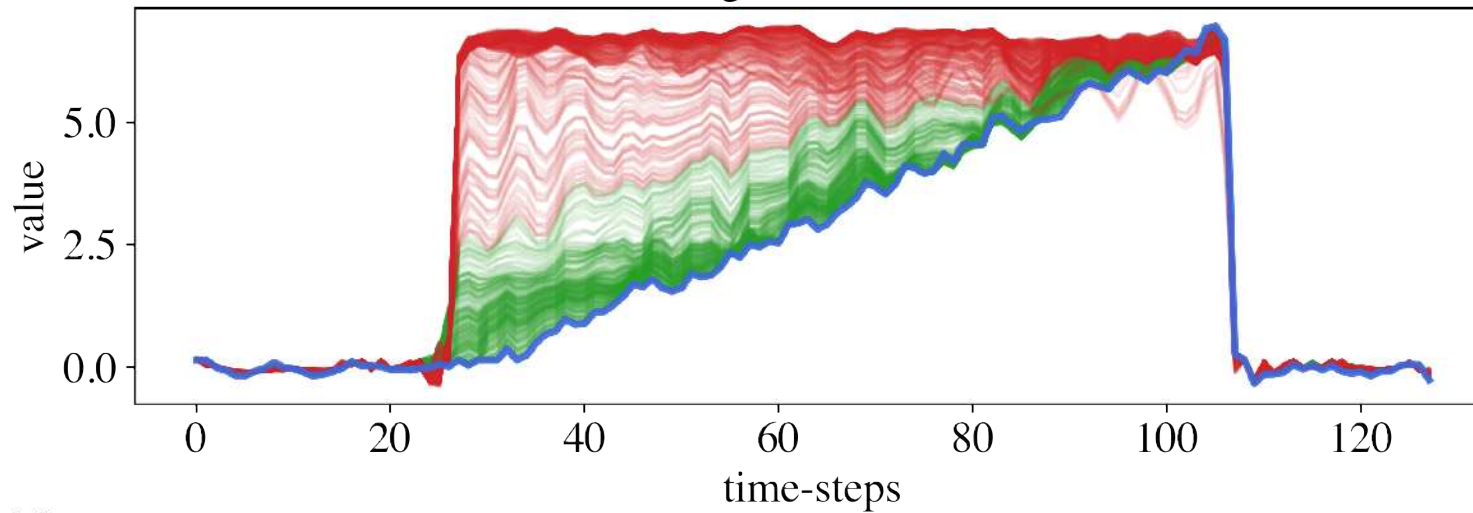
Counterfactuals are instances similar to X and with a different class.



1. Instance-based Explanation

Together they form a local neighborhood.

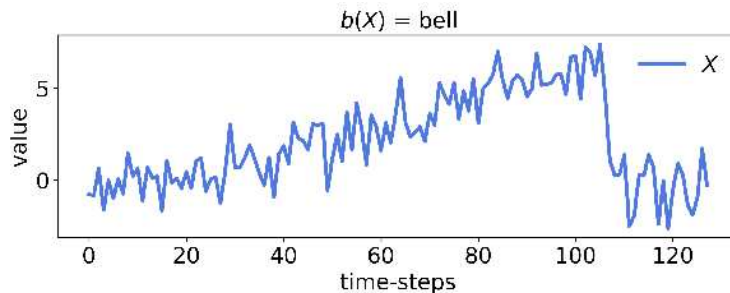
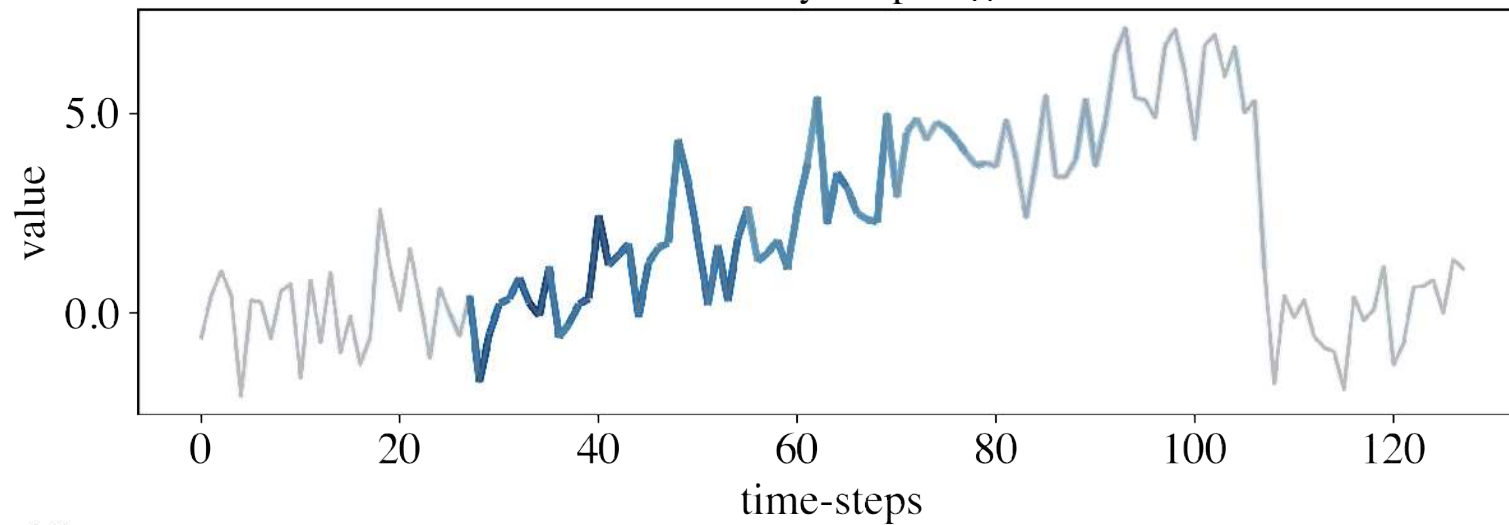
Neighborhood: $\hat{\mathcal{X}}$



2. Element-based explanation

The most *sensitive* areas in the neighborhood form a saliency map.

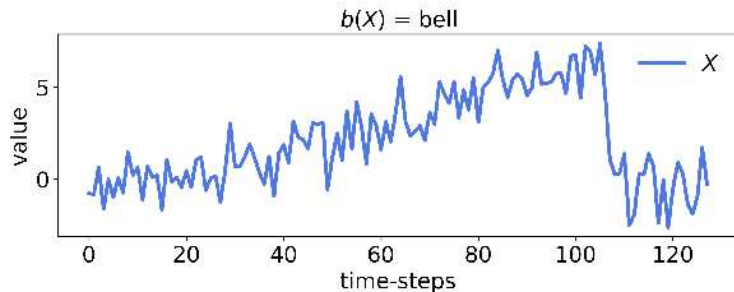
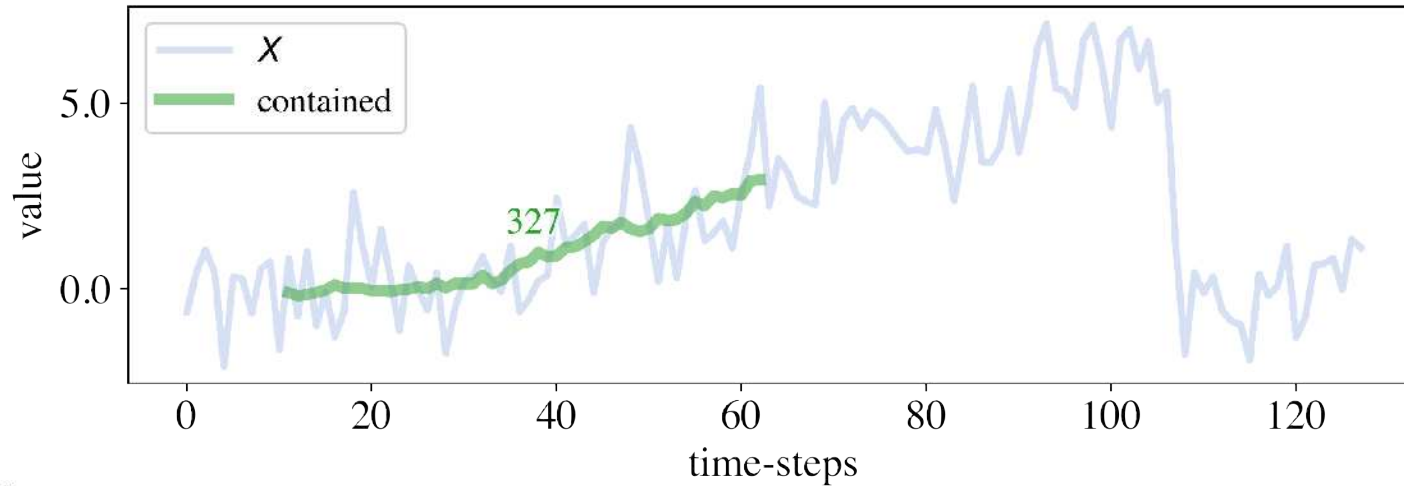
Saliency Map: Φ_X



3. Subsequence-based explanation: Factual Rule

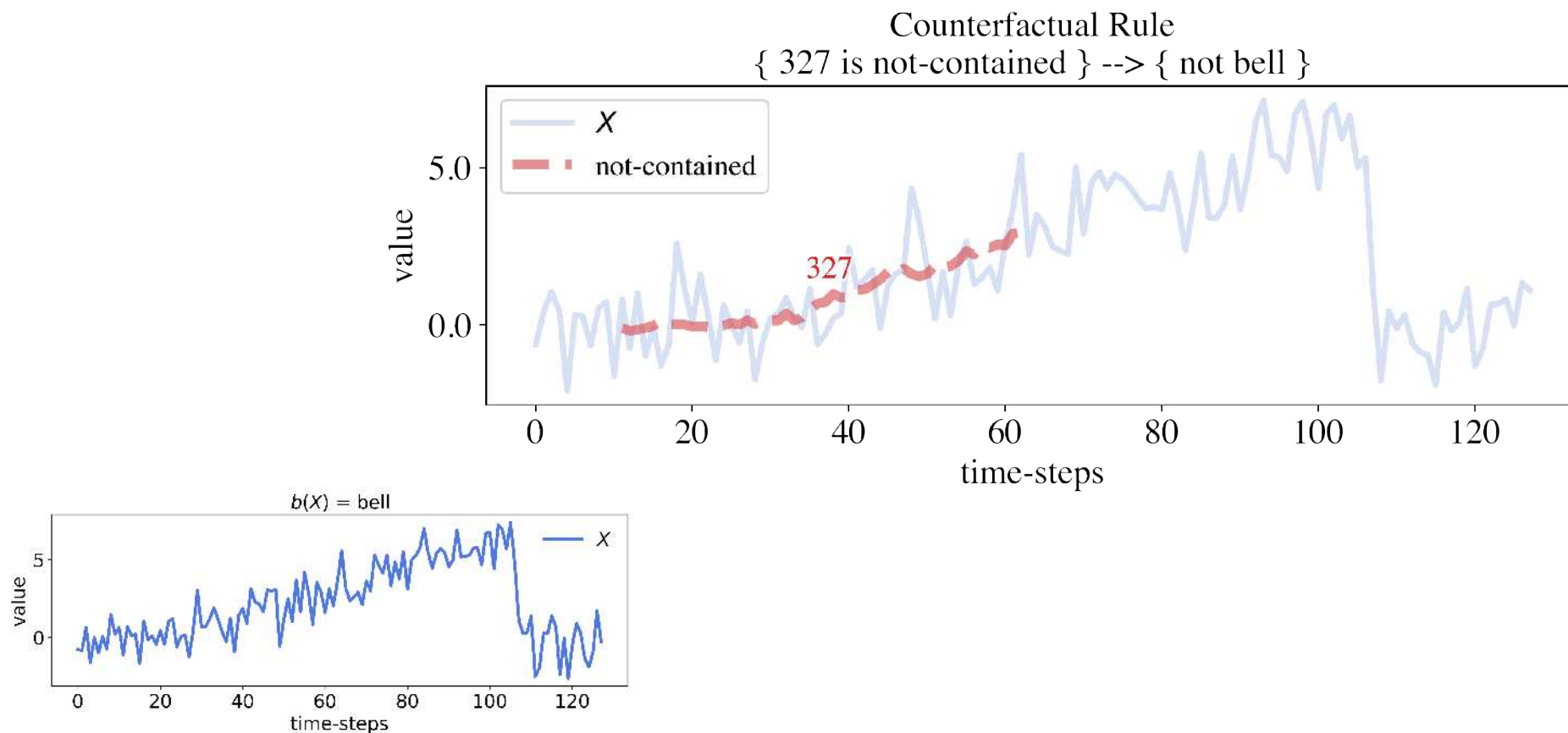
Explains the prediction of the black-box **directly**.

Factual Rule
 $\{ 327 \text{ is contained} \} \rightarrow \{ \text{bell} \}$

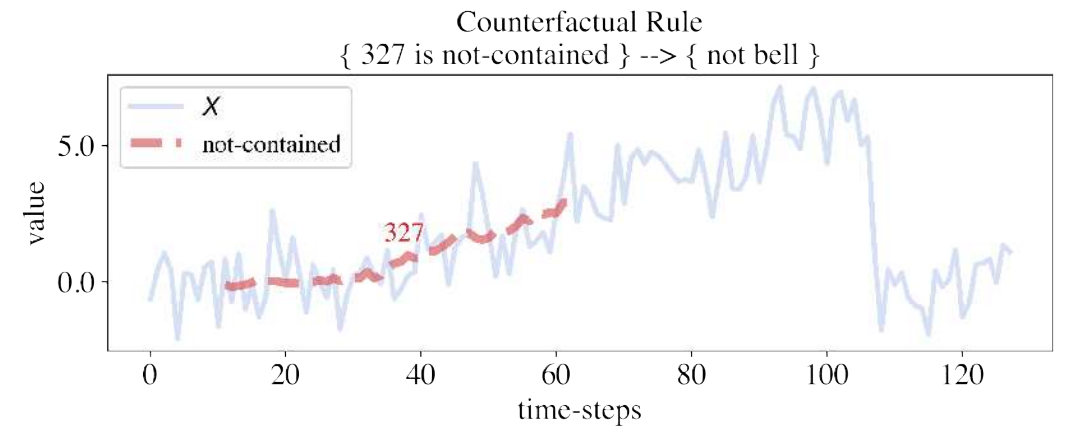
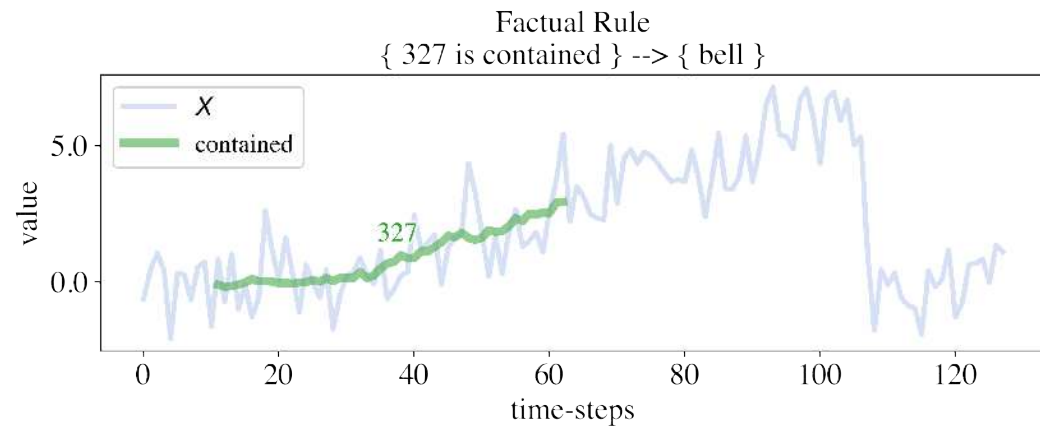
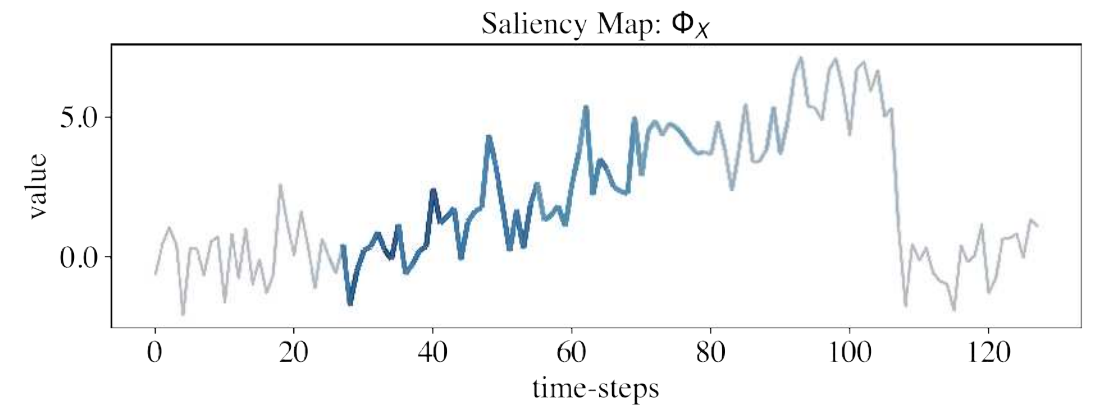
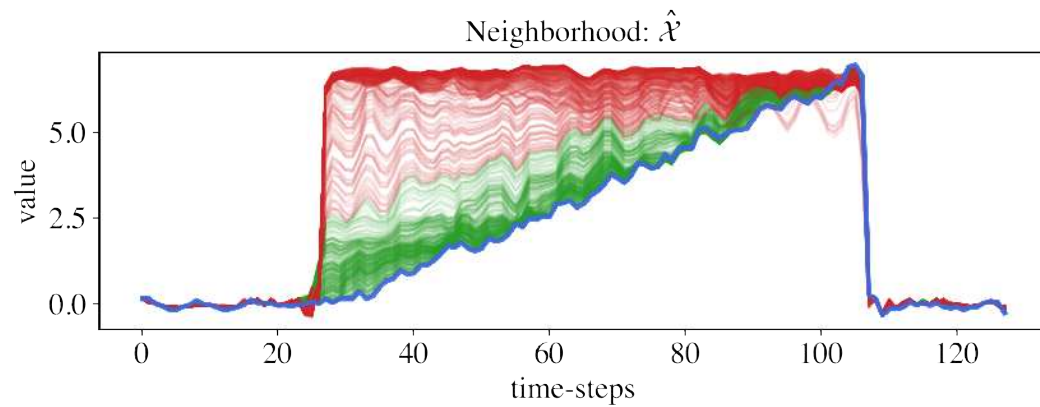


3. Subsequence-based explanation: Counterfactual Rule

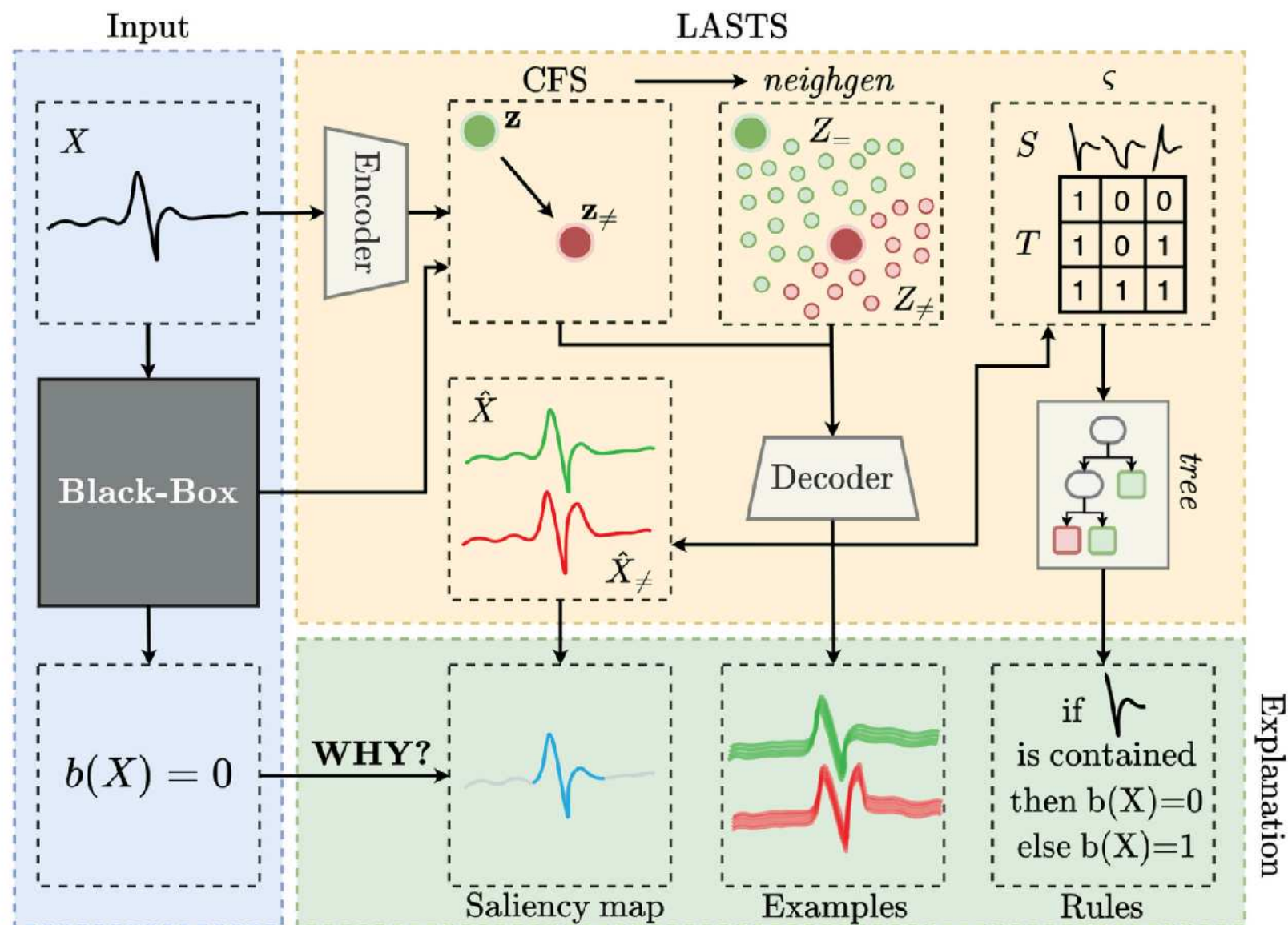
Minimal rule variation resulting in a different black-box prediction.



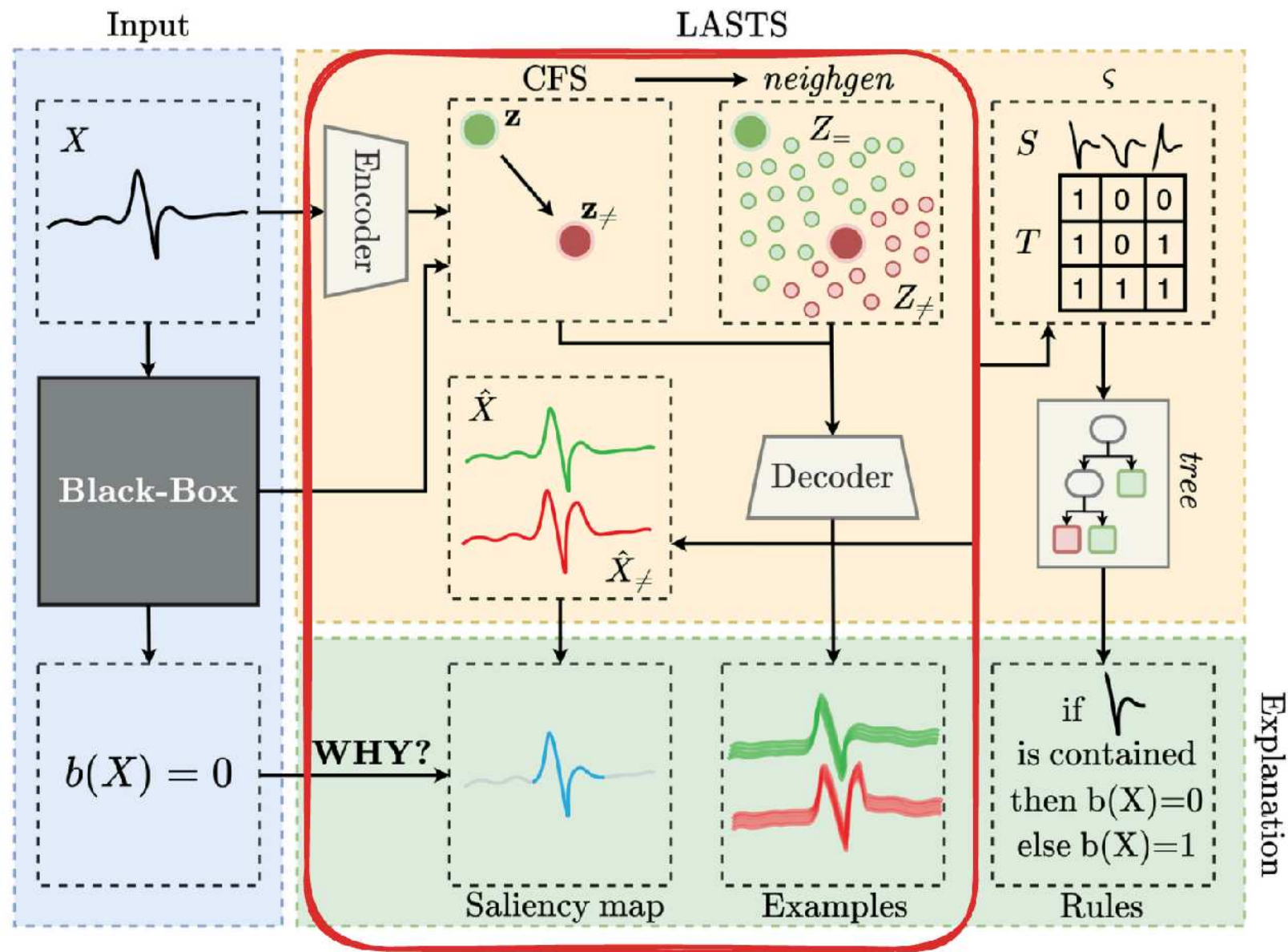
LASTS Explanation



Inside LASTS

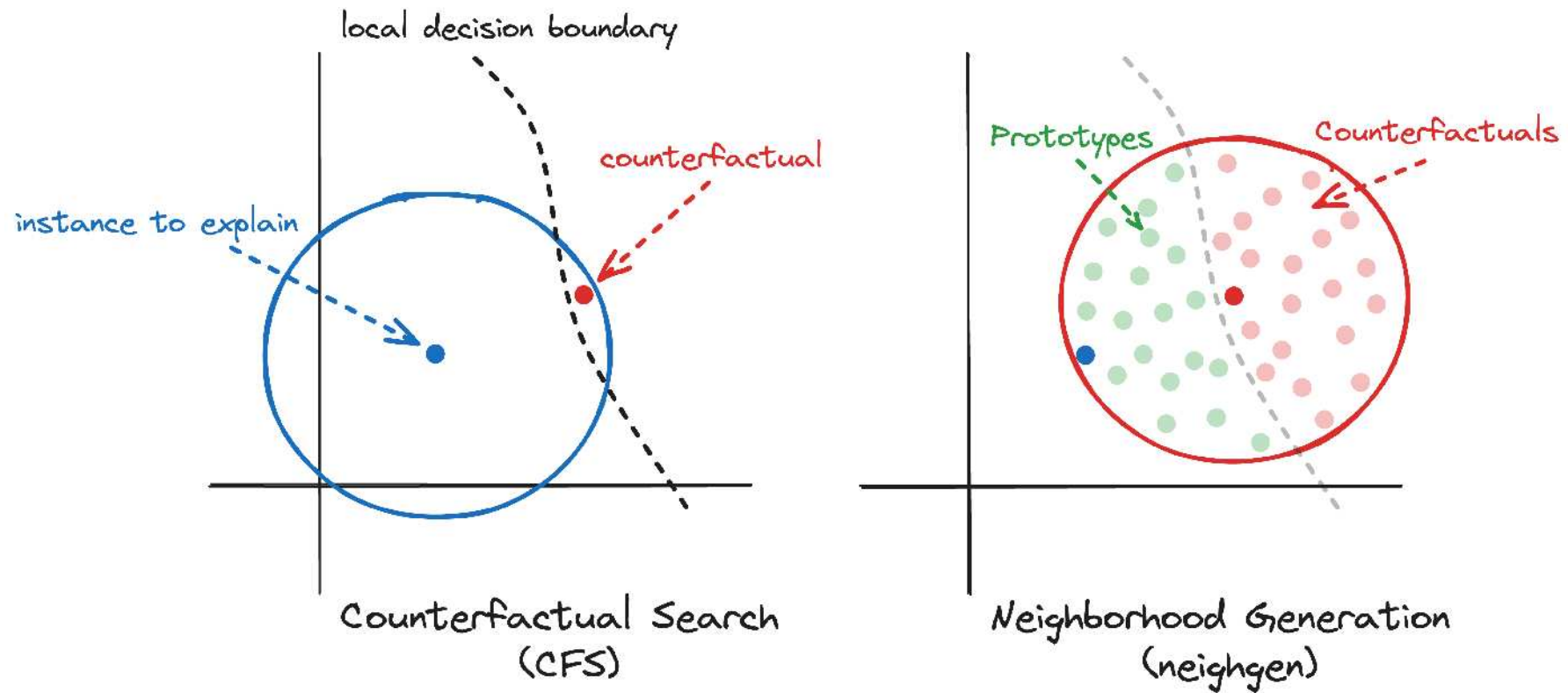


Inside LASTS



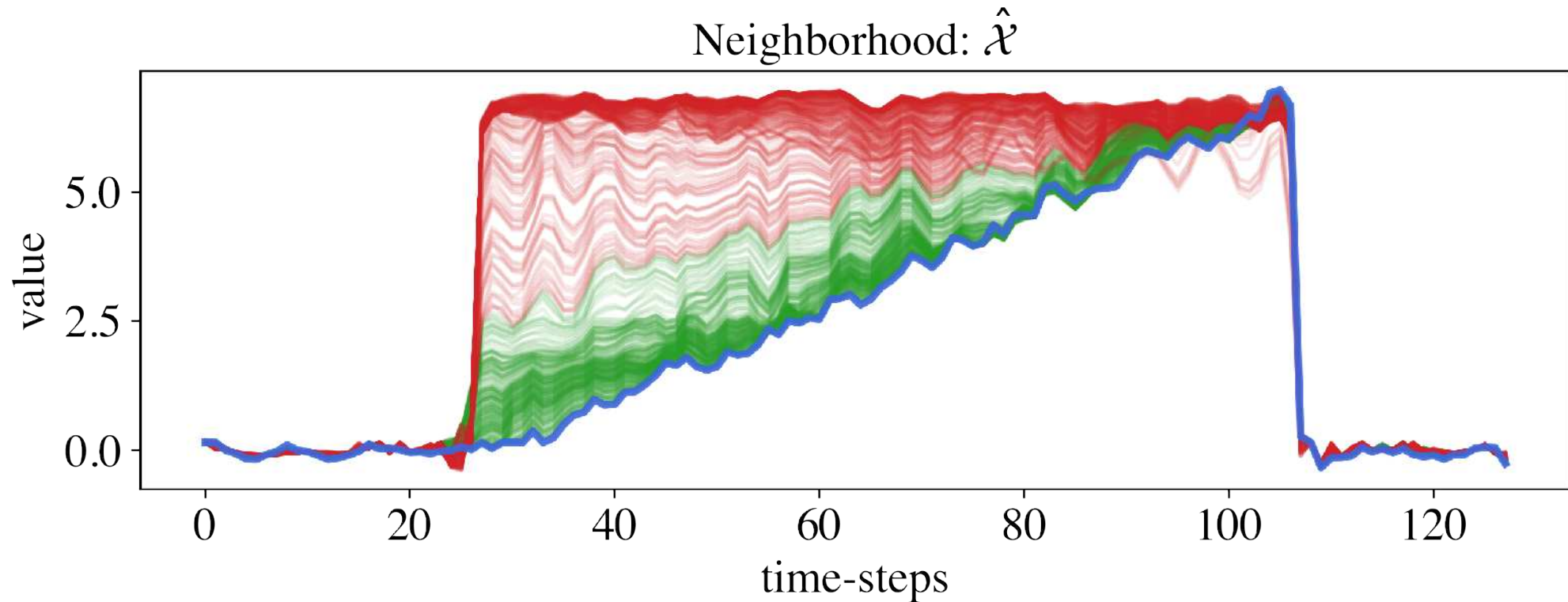
The Latent Space

LASTS uses the latent space of a Variational Autoencoder to **meaningfully** perturb the input time series.



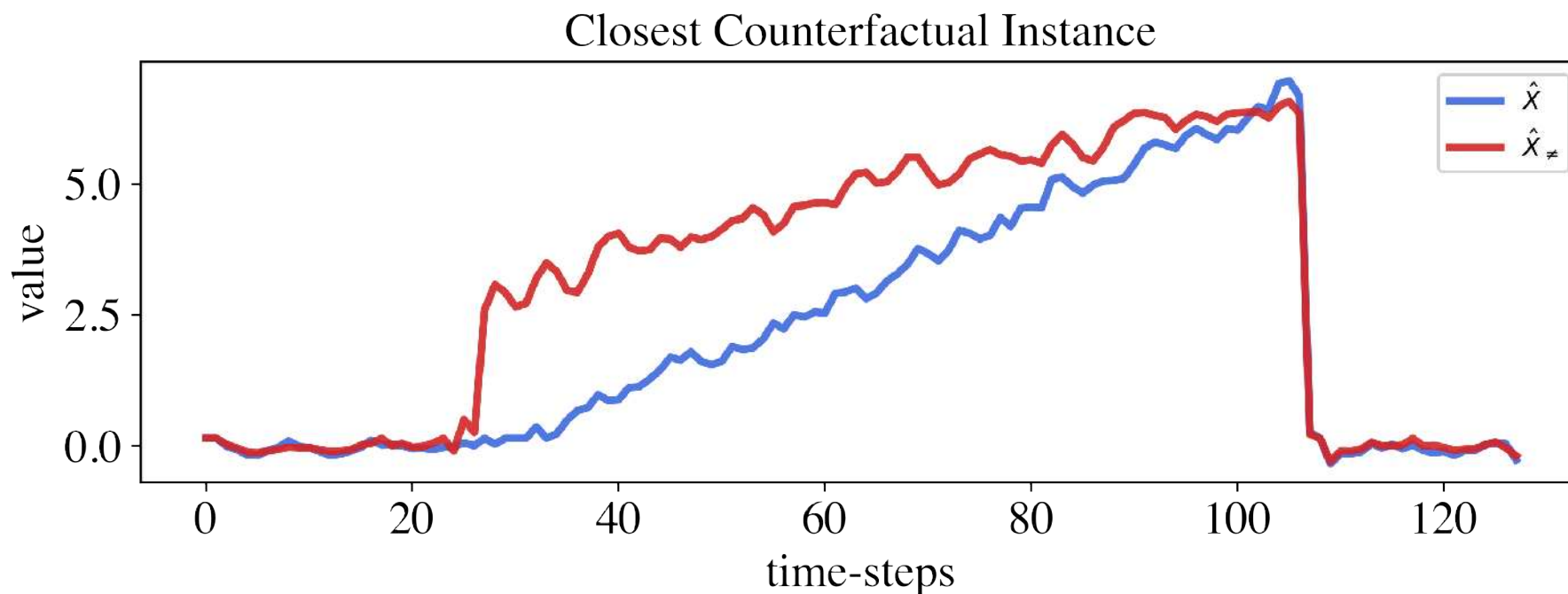
Back to the Time Series Space

LASTS uses the latent space of a Variational Autoencoder to **meaningfully** perturb the input time series.



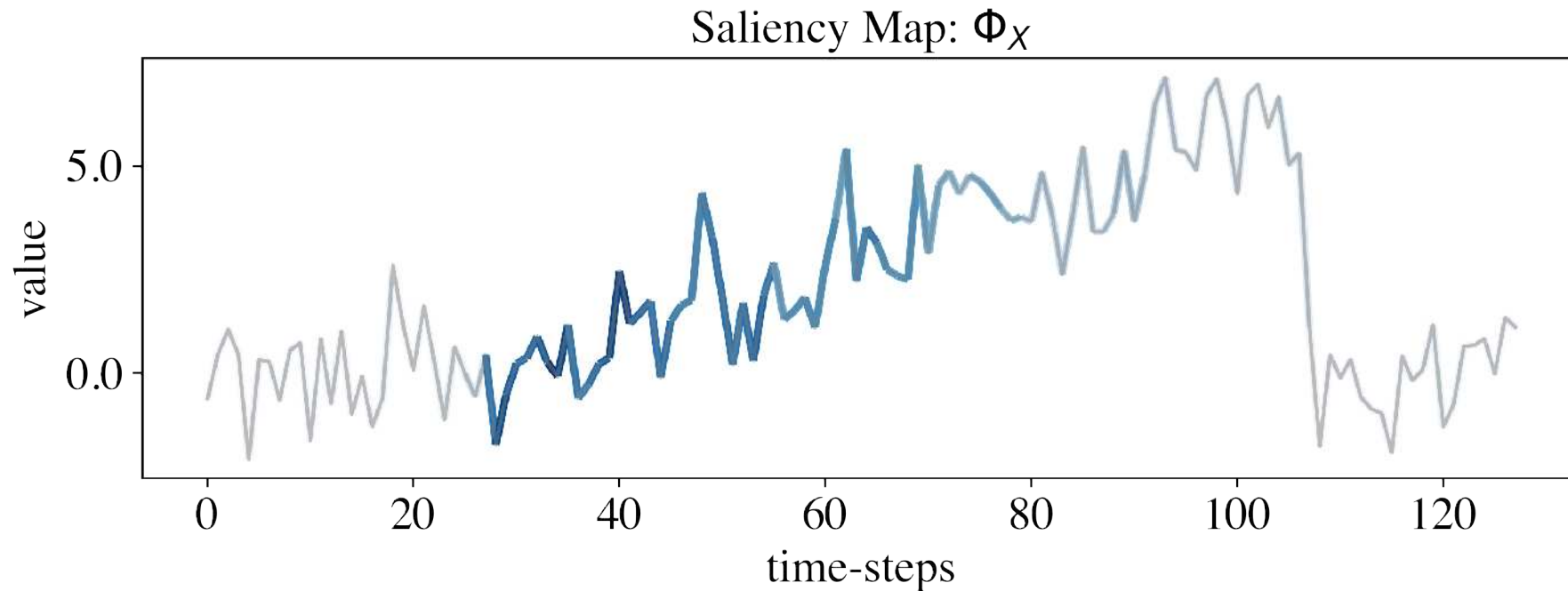
The Closest Counterfactual

The saliency map is obtained by taking **the absolute difference between the instance to explain and the closest counterfactual.**

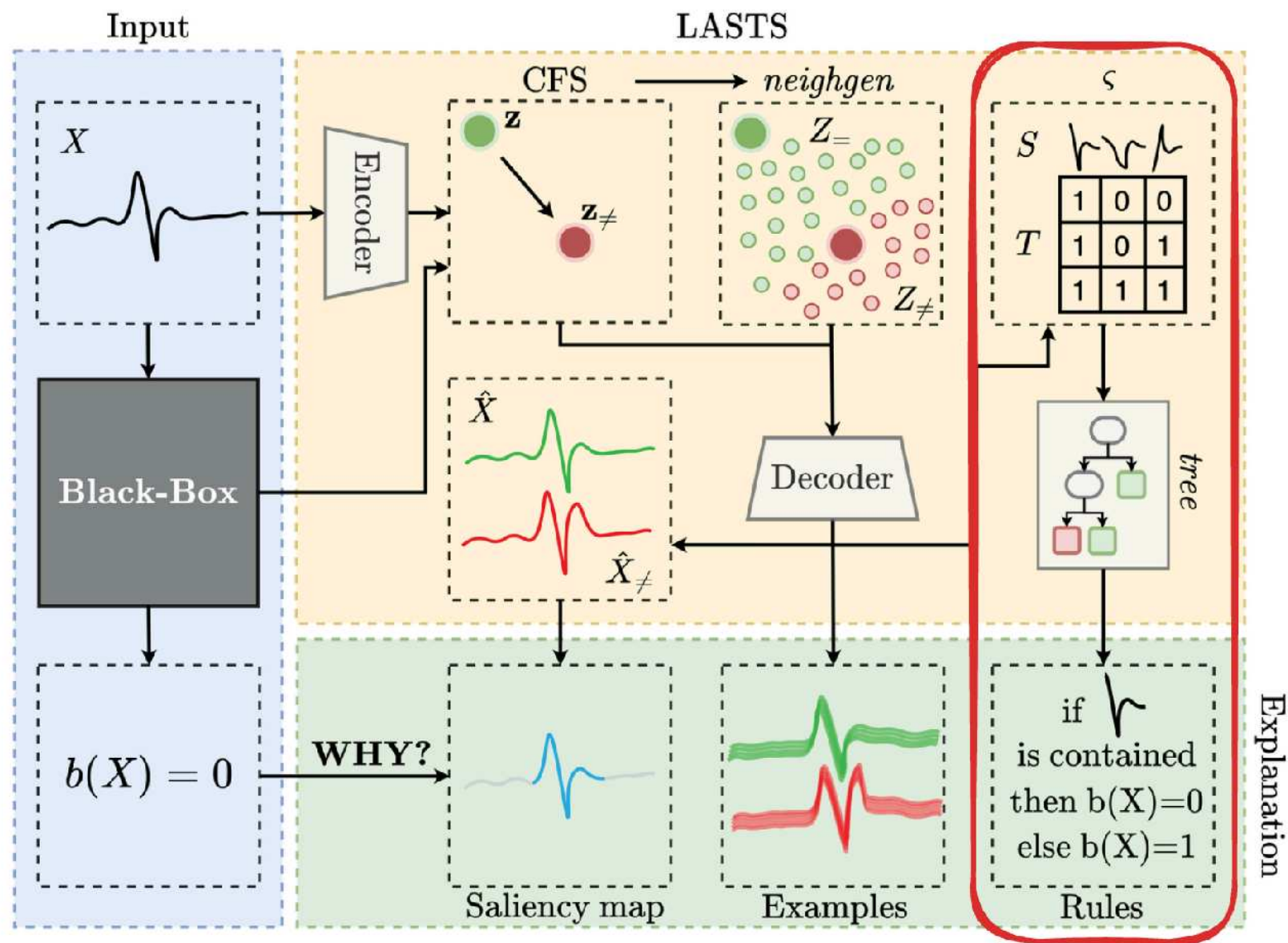


The Saliency Map

The saliency map is obtained by taking **the absolute difference between the instance to explain and the closest counterfactual.**

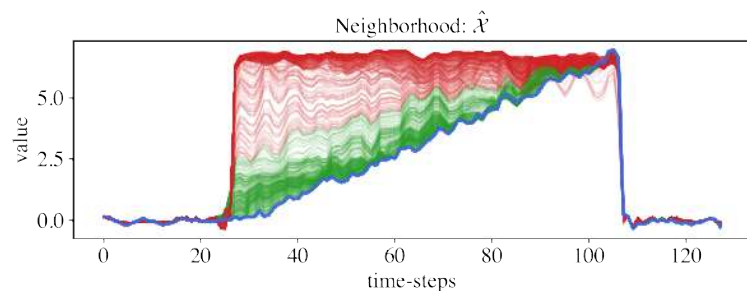


Inside LASTS



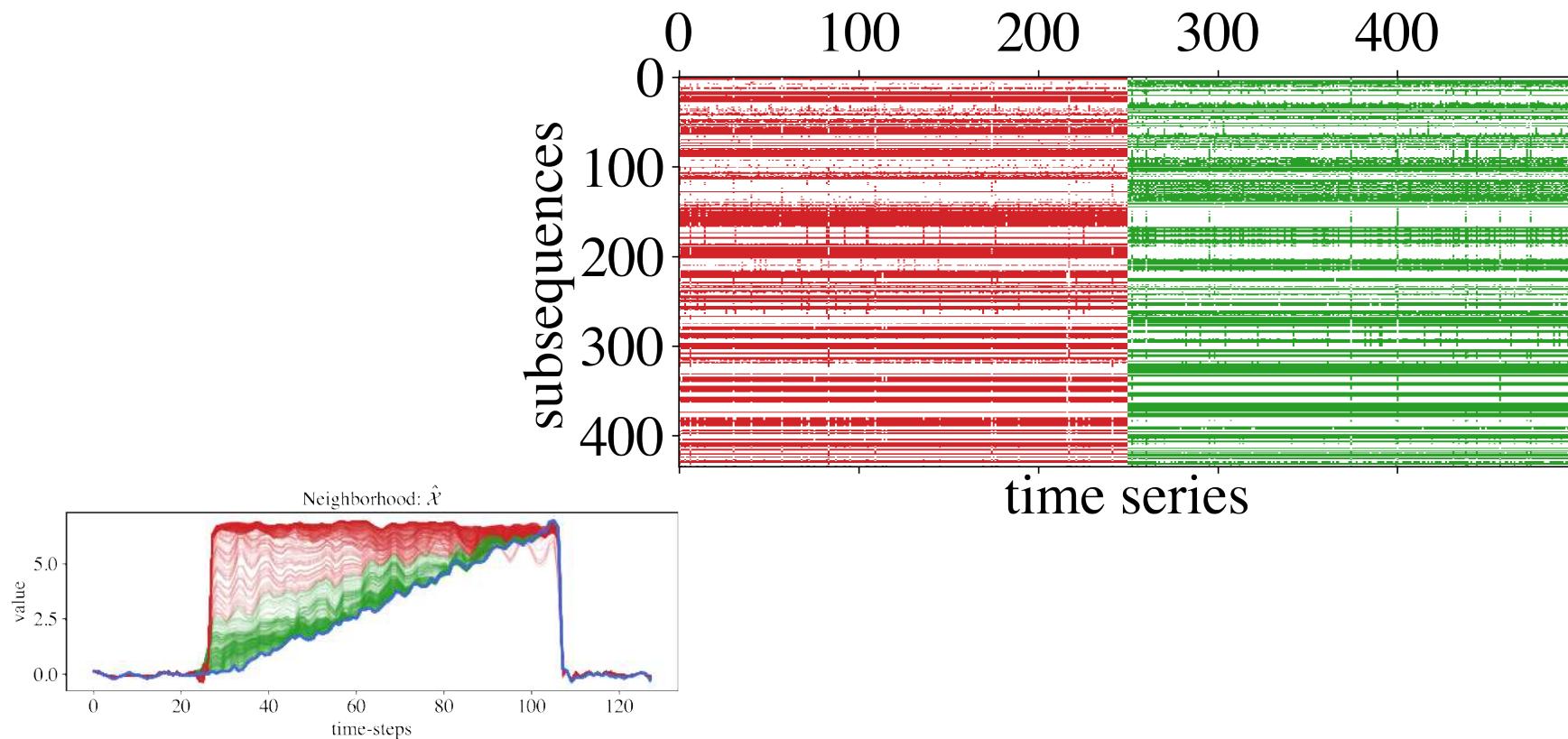
The Interpretable Subsequence-based Surrogate

We convert the synthetic neighborhood in a **subsequence-based representation** with a Bag-Of-Patterns/Shapelet Transform.



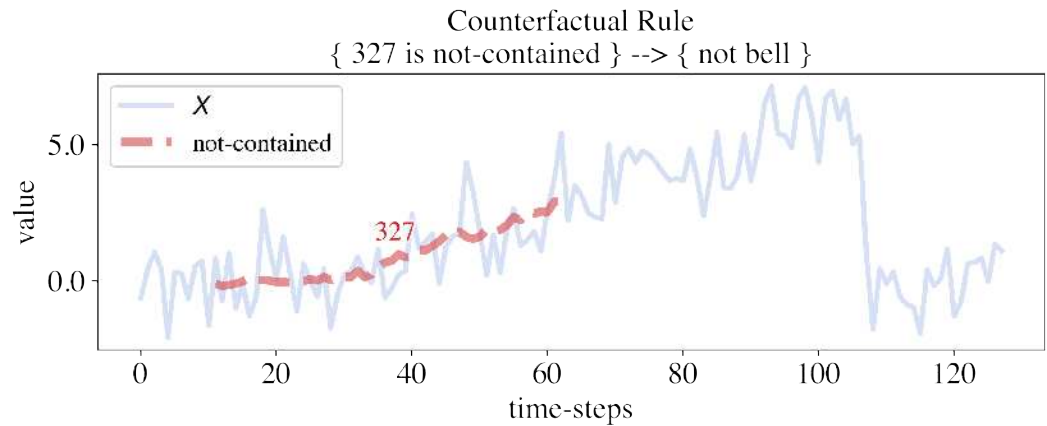
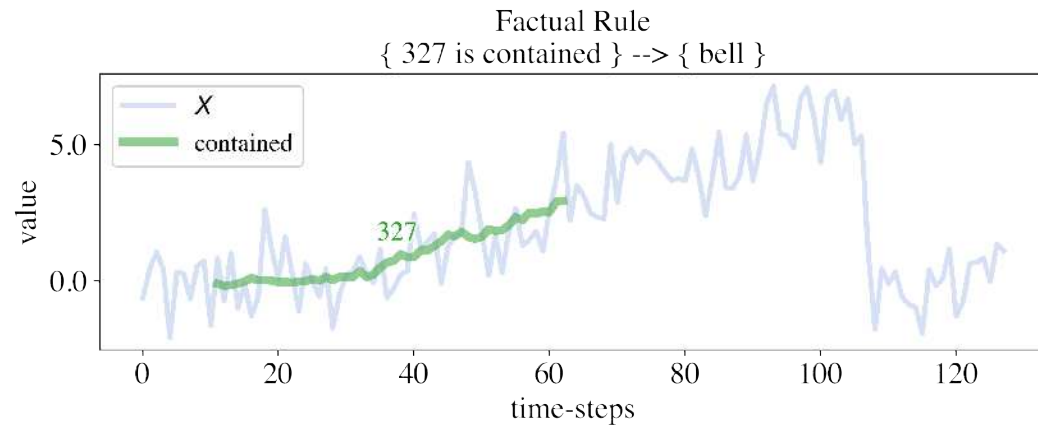
The Interpretable Subsequence-based Surrogate

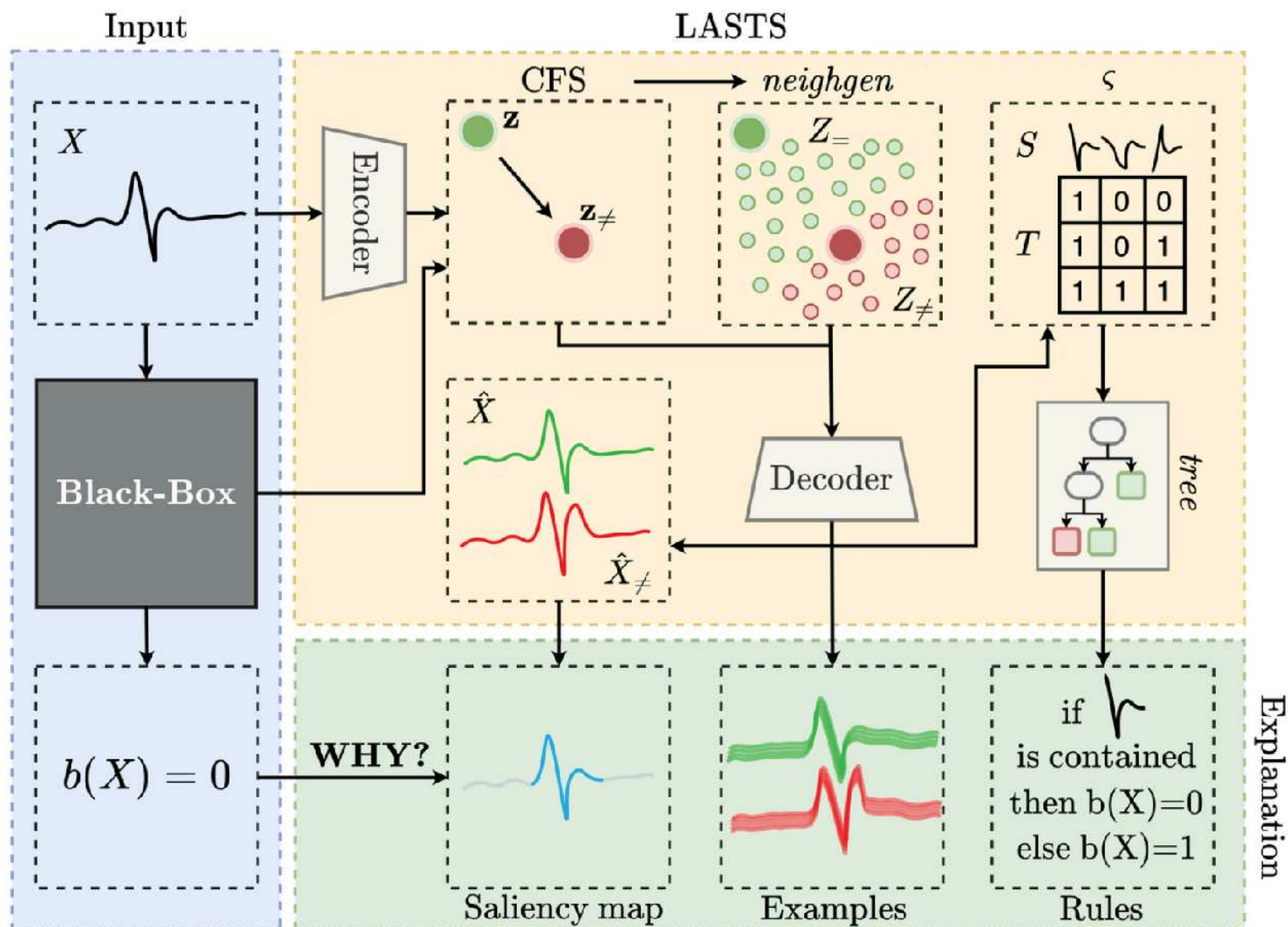
We convert the synthetic neighborhood in a **subsequence-based representation** with a Bag-Of-Patterns/Shapelet Transform.



The Interpretable Subsequence-based Surrogate

The resulting dataset is passed to a **decision tree** from which we extract the factual and counterfactual rules.





Baselines

We compare **different alternatives** for neighborhood generation and subsequence extraction on 4 univariate datasets and 2 black-box models: KNN and ResNet.

As **counterfactual search and neighborhood generation** we test:

- Gaussian (GA) $\rightarrow u \sim \mathcal{N}(0, 1), \quad z_{GA} = z + \theta u$
- Gaussian Matched (GM)* $\rightarrow z_{GM} = z_{GA} / \sqrt{1 + \theta^2}$
- Uniform Sphere (US) $\rightarrow u \sim \mathcal{U}(0, 1), \quad z_{US} = z + \theta u$
- GM search + US generation (MU)

* Agustsson, Eirikur, et al. "Optimal transport maps for distribution preserving operations on latent spaces of generative models." arXiv preprint arXiv:1711.01970 (2017).

We compare **different alternatives** for neighborhood generation and subsequence extraction. on 4 univariate datasets and 2 black-box models: KNN and ResNet.

As **subsequence extraction** methods we test:

- Shapelet Transform^{*}
- SAX-based Bag-Of-Patterns^{**}

^{*} Grabocka, Josif, et al. "Learning time-series shapelets." Proceedings of the 20th ACM SIGKDD international conference on Knowledge discovery and data mining. 2014.

^{**} Le Nguyen, Thach, et al. "Interpretable time series classification using linear models and multi-resolution multi-domain symbolic representations." Data mining and knowledge discovery 33 (2019): 1183-1222.

Baselines

The best approach in terms of **fidelity** uses a SAX-based Bag-Of-Patterns and MU search/neighborhood generation.

	MODEL	GEN-SHP	GA-SHP	GA-SAX	GM-SHP	GM-SAX	MU-SHP	MU-SAX	US-SHP	US-SAX	GLO-SHP	GLO-SAX
CBF	RES	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	0.94
	KNN	1.00	1.00	1.00	0.97	1.00	1.00	1.00	1.00	1.00	1.00	0.97
COF	RES	1.00	0.93	0.93	0.93	1.00	0.93	1.00	1.00	1.00	0.75	0.89
	KNN	1.00	0.93	1.00	0.93	1.00	0.96	1.00	1.00	1.00	0.89	0.89
EC2	RES	0.98	0.94	1.00	0.96	0.99	0.98	1.00	0.95	0.99	0.69	0.78
	KNN	0.97	0.96	0.97	0.90	0.99	0.95	0.98	0.97	0.98	0.83	0.83
GUN	RES	0.98	0.82	0.98	0.78	0.98	0.84	0.98	0.80	0.98	0.93	0.90
	KNN	0.96	0.86	0.96	0.82	0.98	0.86	0.98	0.86	0.98	0.74	0.81
$\overline{rk}$		4.38	7.31	4.44	8.50	3.31	6.69	3.25	5.88	3.50	8.81	9.94

State-Of-The-Art Comparison

We benchmark the best version of LASTS on **15 datasets**, 10 univariate and 5 multivariate for the UEA/UCR repositories, using **ROCKET*** as black-box.

Each part of the explanation returned by LASTS is evaluated with different metrics and benchmarks.

* Dempster, Angus, François Petitjean, and Geoffrey I. Webb. "ROCKET: exceptionally fast and accurate time series classification using random convolutional kernels." Data Mining and Knowledge Discovery 34.5 (2020): 1454-1495.

State-Of-The-Art Comparison

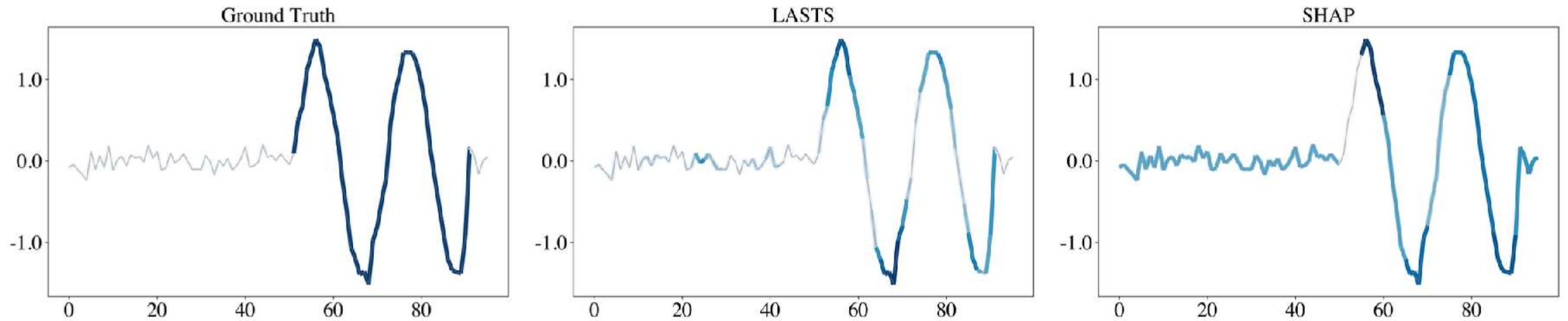
- instance-based → usefulness.
- saliency-based (against SHAP*) → stability, correctness and insertion/deletion benchmarks.
- rule-based (against a global surrogate and ANCHOR**) → fidelity, precision and coverage.

* Scott, M., and Lee Su-In. "A unified approach to interpreting model predictions." Advances in neural information processing systems 30 (2017): 4765-4774.

** Ribeiro, Marco Tulio, Sameer Singh, and Carlos Guestrin. "Anchors: High-precision model-agnostic explanations." Proceedings of the AAAI conference on artificial intelligence. Vol. 32. No. 1. 2018.

SHAP Comparison

We use a **synthetic experiment** to check if the saliency maps obtained with agnostic explainers match with **custom-defined ground truths** (synthetic classifier on synthetic data).



Limitations and Future Works

The autoencoder is a strength, but also a weakness because it is a black-box and **has to be trained for each dataset**. But:

- it could be replaced by recent, more powerful models like **diffusion models**;
- it would not require re-training with **transfer learning** if there was a **foundation model** for time series data.

The **time series regression** task could be interesting, but there is no decision boundary to **define prototypes and counterfactual**.

THANK YOU FOR THE ATTENTION!

 github.com/fspinna/lasts



SOBIGDATA
RESEARCH INFRASTRUCTURE



Future
Artificial
Intelligence
Research

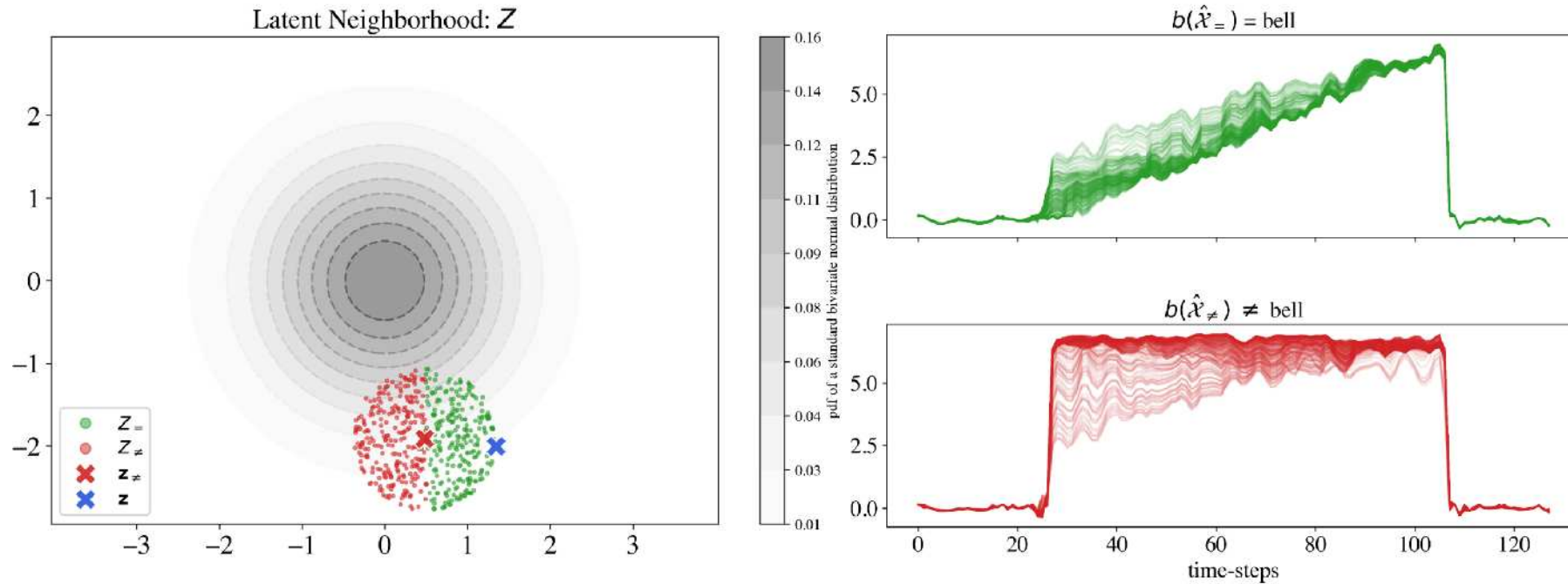
HUMANE



AI NET

Bonus Slide: The Latent Space

We aim to sample synthetic time series **consistent** to the dataset, i.e., from *probable* parts of the space, given a normal distribution.

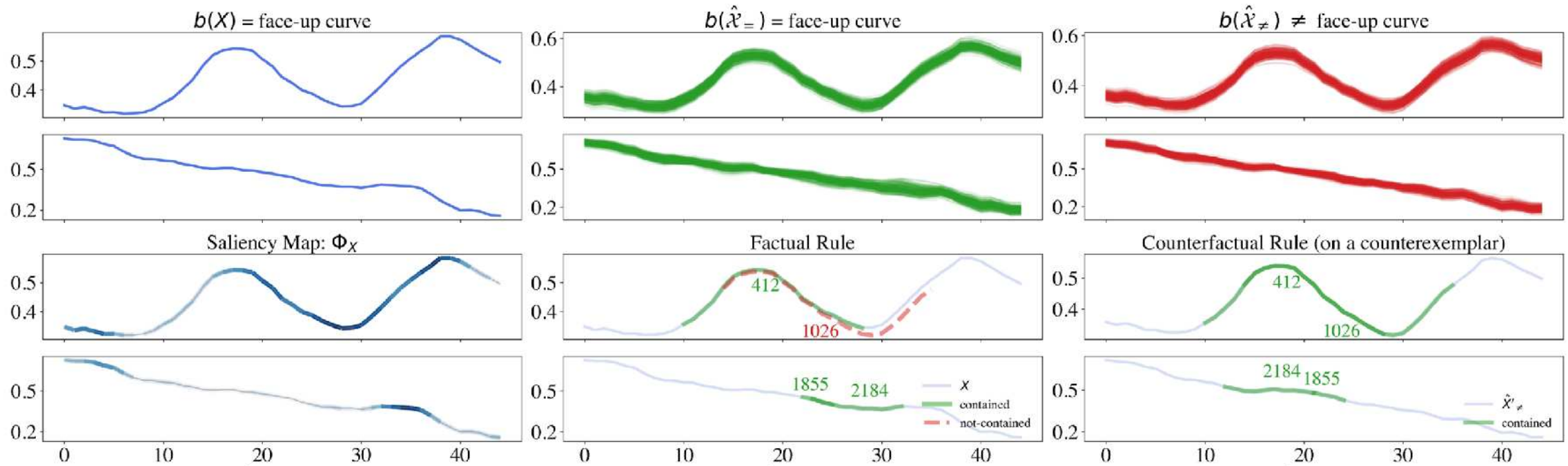


Bonus Slide: Baselines

This version is also much faster than the previous version of LASTS.

		<i>CBF</i>	<i>COF</i>	<i>EC2</i>	<i>GUN</i>	
<i>neighgen</i>	CFS	4.83 ± 0.18	8.18 ± 0.44	6.81 ± 0.54	7.47 ± 0.63	
	GENETIC	55.57 ± 4.55	80.67 ± 2.11	72.25 ± 5.14	80.81 ± 18.00	
ζ^*	SHP	60.02 ± 5.43	46.19 ± 5.24	47.33 ± 8.53	42.25 ± 7.85	
	SAX	8.91 ± 0.47	43.75 ± 9.33	13.04 ± 2.61	12.63 ± 2.14	
<i>all</i>	GEN-SHP	115.59 ± 9.98	126.86 ± 7.35	119.58 ± 13.67	123.06 ± 9.83	OLD
	MU-SHP tot	64.85 ± 5.61	54.38 ± 5.69	54.14 ± 9.07	49.72 ± 8.48	
	MU-SAX tot	13.74 ± 0.65	51.93 ± 9.77	19.85 ± 3.15	20.10 ± 2.77	NEW!

Bonus Slide: Explaining Libras



Bonus Slide: Explaining ECG5000

